

Week 6 — Activity: Visualizing Magnetic Fields

Geophysics 3A: Potential Fields & Geothermics

April 10, 2026

Purpose

This activity develops physical intuition for magnetic fields as vector fields generated by dipoles. You will explore how field geometry depends on source orientation, strength, and proximity.

Materials

- Disc magnets
- Paper sheets
- Iron filings

Part A: Single disc magnet — vertical orientation

Place a disc magnet flat on the table and sprinkle iron filings over a sheet of paper on top.

1. Describe the symmetry of the field pattern.
2. Where are the field gradients strongest?
3. If this magnet were buried at depth, where would you expect the strongest anomaly?

Sketch:

Part B: Single disc magnet — horizontal orientation

Stand the same magnet on its edge and repeat the experiment.

1. How does the field symmetry differ from Part A?
2. Where are positive and negative lobes located?
3. How would this differ from a gravity anomaly produced by the same body?

Sketch:

Part C: Stacked disc magnets

Stack several disc magnets and repeat Parts A and B.

1. How does stacking affect field strength?
2. Does stacking change anomaly shape or only amplitude?
3. What geological parameter does stacking mimic?

Reflection

Why are magnetic field patterns sensitive to orientation, while gravity fields are not?